

Training the model

SETTING HYPERPARAMETERS

Idea #1: choose the hyperparameters that work best on *the data*.

BAD: the network can memorize the training set.



Idea #2: split into **train** and **test**; choose what works best on test.

BAD: no idea how it will perform on new data.



Idea #3: split into **train**, **val** and **test**; tune on val, evaluate once on test.

Better!



tune here

open once, at the end

A typical split is 60 / 20 / 20. The exact numbers matter far less than never tuning on test.

SETTING HYPERPARAMETERS: CROSS-VALIDATION

Your dataset

Idea #4: cross-validation. Split the data into **folds**, try each fold as validation, and average the results.

fold 1	fold 2	fold 3	fold 4	fold 5	test
fold 1	fold 2	fold 3	fold 4	fold 5	test
fold 1	fold 2	fold 3	fold 4	fold 5	test

Widely used in small datasets

The medical mistake that voids a study

Split by **patient**, never by image. Folds included.

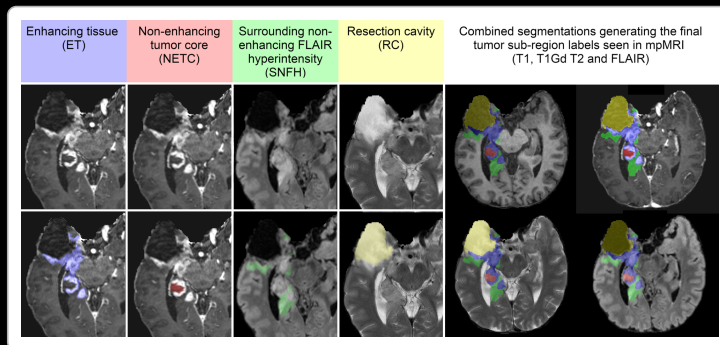
Two slices of the same tumor in train and test do not measure generalization; they measure memory. This inflates accuracy quietly and is the single most common flaw in published medical deep learning.

When labels are scarce

LABELS ARE HARD TO OBTAIN

Expensive

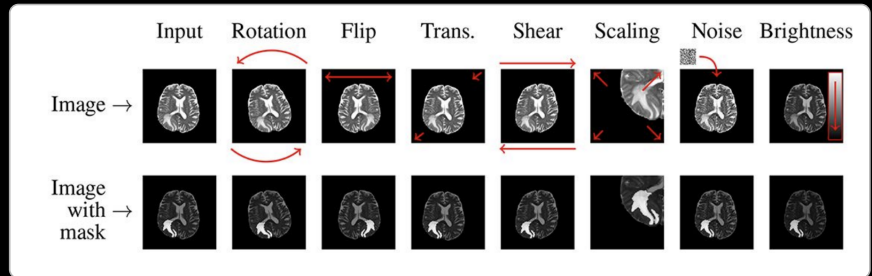
A label needs a **specialist's** time. The 2024 BraTS post-treatment glioma challenge pooled $\sim 2,200$ cases from seven hospitals in two countries, and the scans were the cheap half.



Each case: four MRI sequences, four tumour sub-regions, **drawn voxel by voxel** in 3D: refined by a clinical annotator, then signed off by a board-certified neuroradiologist, and returned until it passed.

STRATEGY 1: MAKE YOUR DATA GO FURTHER, WITH AUGMENTATION

Apply affine and pixel transformations to your data → more training samples.



Why it works

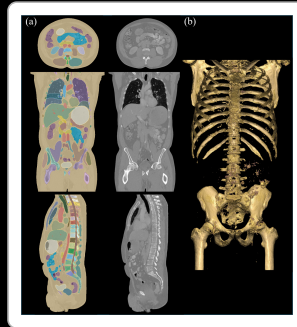
Each transform is a statement about what the label is *invariant* to. You are handing the network a prior for free.

Note the second row

A geometric transform must be applied to the **mask** as well. Get this wrong and you train on silently mislabeled data.

Can we automate this?

STRATEGY 2: GENERATE THEM, WITH SYNTHETIC IMAGES



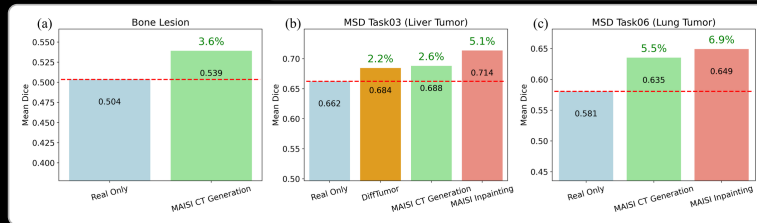
The conditioning organ mask, and the CT it produced.

MAISI

A **latent diffusion** model for 3D CT. It denoises in a compressed latent space, and a ControlNet conditions the output on an organ mask of 127 anatomical structures, volumes up to $512 \times 512 \times 768$.

More than augmentation

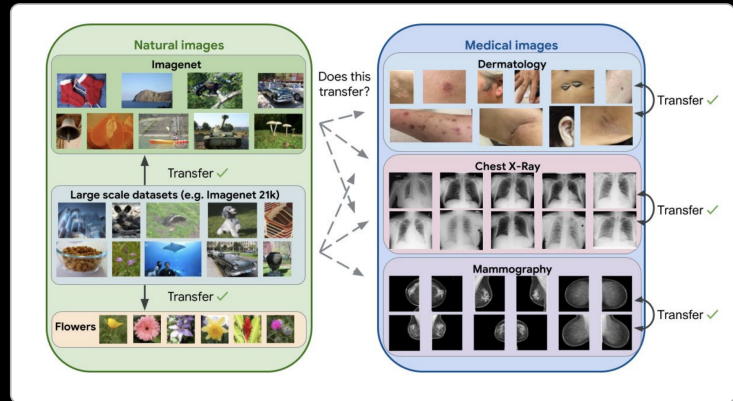
You hand it the mask, so **the label is generated with the image**: a new labelled example, not a re-photographed one. The catch: synthetic anatomy can be confidently wrong. Fine for training, **never** for diagnosis.



Generated volumes mixed into real training data lift tumour segmentation Dice on every task tried; green is the gain over real data alone (three of six panels). We build one properly in Lecture 3.

STRATEGY 3: BORROW FROM ANOTHER DOMAIN, WITH TRANSFER LEARNING

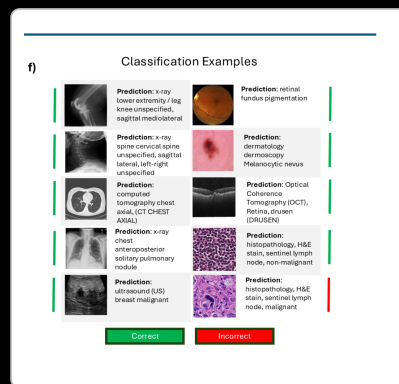
Pretrain on a large natural-image dataset, then fine-tune on the medical task. **Does that transfer?**



Largely yes. Early filters (edges, textures, gradients) are not specific to cats, and dermatology, chest x-ray and mammography all benefit. Later layers are where the domains genuinely diverge, which is why we fine-tune those.

STRATEGY 4: NEED AN OUT-OF-THE-BOX TOOL?

MedImageInsight, an open-source medical imaging embedding model. Trained on medical images with associated text and labels across a diverse collection of domains, including X-ray, CT, MRI, dermoscopy, OCT, fundus photography, ultrasound, histopathology, and mammography.



Classification results, zero-shot, no fine-tuning:

Dataset	Domain	mAUC
IRMA	X-ray	0.989
BUSI	Ultrasound	0.985
ISIC2019	Dermoscopy	0.966
PCAM	Histopathology	0.963
OCT2017	OCT	1.000
RSNA BoneAge	X-ray (age)	6.19 mo

■ exceeds human expert ■ SOTA equivalent ■ AUC > 0.9

< **0.61B** total parameters

14 imaging domains